

Week 1 Bachelor

What I Have Learned From Each Paper

Survey of Hallucination in Natural Language Generation

- Definition of hallucination and its types (Intrinsic and Extrinsic)
- Hallucination metrics for LLMs (Reference-dependent and Reference-free)
- Hallucination Mitigation : how to fix hallucinations in LLM using different approaches

A Survey on Hallucination in Large Language Models

- Modern definition and categorization of hallucination in LLM
- Causes of hallucinations
- Hallucination detection (Factuality and faithfulness hallucination detection)
- Hallucinations benchmarks (Evaluation and detection benchmarks)
- Hallucination mitigation and how to try to fix the hallucinations using different ways (Data related, Training related and Inference related)
- Hallucination in RAG because of user queries, retrieval source and retriever

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

- RAG models (RAG-Sequence and RAG-Token)
- Retriever: DPR-based
- Generator: BART
- Experiment of RAG with different approaches

SelfCheckGPT

- Grey-Box factuality assessment and how it is used to detect hallucination using next word probability
- Black-Box factuality assessment and how it uses proxy LLM to detect hallucinations
- SelfCheckGPT with BERTScore by comparing sentences in LLM response with all sentences in samples generated with same prompt used for the response
- SelfCheckGPT with QA and how to use the MQAG framework and compare answers across response and samples
- SelfCheckGPT with n-gram by training n-gram model on samples and responses then compute the avg log probability of sentences in response
- SelfCheckGPT with NLI by using BeBERT-V3 large fine tuned MNLI as the NLI-model and concatenate the promise (Samples s^n) and hypothesis (sentence c_i) then normalize and then compute the avg contradiction across all samples
- SelfCheckGPT with prompt by providing the llm prompt with context (sample) and sentence and the question and the answer format

Adaptable Logical Control for Large Language Models

- Ctrl-G is a scalable and reliable framework to control LLM to follow logical constraints
- The three major steps of Ctrl-G
- Distillation to train HMM as white box
- Construct DFA present desired logical constraints
- Inference: to condition HMM on DFA

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Gaps in Previous Work

- Hallucination in large language Multimodal
- hallucinations in long tail and low resource domains
- understanding hallucinations in LLM
- challenge of delineating and understanding knowledge boundaries within LLMs
- The retriever and generator in RAG are trained separately, then combined. Future work could explore joint pre-training of both components from scratch
- SelfCheckGPT : 1-Limited domain scope: The study only evaluated hallucinations in biographies. Future work should test on a wider range of concepts (e.g., locations, objects) to see if findings generalize. 2-Coarse-grained factuality detection: The method operates at the sentence level, but sentences can contain both factual and non-factual information. Future work should adopt fine-grained evaluation by decomposing sentences into atomic facts (Min et al., 2023). 3-High computational cost: The best-performing variant (SelfCheckGPT-Prompt) is computationally expensive, limiting practical use. Future work should focus on making it more efficient without sacrificing accuracy.
- Implementation for Ctrl-G is naive and it left for future research for better algorithms used

Week 1 Bachelor

What To Do Next

- symbolic reasoning with AI
- how to define domain knowledge
- different approaches to detect hallucinations
- More ways to make LLM follow logical constraints
- how to make effective prompts and how LLM can hallucinate from prompts
- RAG more modern approaches